AN OPEN COVERING CAN BE REFINED BY MEMBERS OF ANY BASIS

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *Let X be a topological space. Let $\mathcal{B}$ be a basis for the topology on X. Let $\mathcal{U} : U = \bigcup_i U_i$ be an open covering of $U \subset X$. There exists an open covering $U = \bigcup V_j$ which is a refinement of $\mathcal{U}$ such that each V_j is an element of the basis $\mathcal{B}$.*

Proof. If $x \in U = \bigcup_{i \in I} U_i$, there is an $i_x \in I$ such that $x \in U_{i_x}$. Thus we have a $B_{i_x} \in \mathcal{B}$ verifying $x \in B_{i_x} \subset U_{i_x}$. Set $J = \{i_x | x \in U\}$ and for $j = i_x \in J$ set $V_j = B_{i_x}$. This gives the desired open covering of U by $\{V_j\}_{j \in J}$. □

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